



Three-dimensional multiphase modeling of cold start processes in polymer electrolyte membrane fuel cells

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ABSTRACT

Startup from subzero temperatures, referred to as “cold start”, has been one of the technical challenges hindering the commercialization of polymer electrolyte membrane fuel cell (PEMFC). In this study, a mathematical model has been developed to simulate the cold start processes in a PEMFC. The present three-dimensional multiphase model uniquely includes the water freezing in the membrane electrolyte, the non-equilibrium mass transfer between the water in the ionomer and the water (vapour, liquid and ice) in the pore region of the catalyst layer (CL), and the water freezing and melting in the CL and gas diffusion layer (GDL). Both the failed and successful cold start processes have been numerically investigated. Numerical results indicate that increasing the ionomer fraction in the cathode CL has more significant effects than increasing the thickness of the membrane layer in reducing the amount of ice formation, and the ohmic heat is the largest heating source at low cell voltages. It is observed that water freezes first in the cathode CL under the land, and ice melts first in the CLs under the flow channel, the melted water in the anode is also removed faster than in the cathode.

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1. Introduction

Low operating temperature, high power density and zero/low emission have made polymer electrolyte membrane fuel cell (PEMFC) the most promising fuel cell type for the future in automotive applications. It appears to offer the best prospect for further improvement until commercialization. Over past decades PEMFC technology has achieved significant improvements in terms of performance, durability, stability and cost. However, further improvement is necessary before commercial application, such as cold start in winter. In winter conditions, it is unavoidable for vehicles driving below the freezing point of water (0°C), therefore, for successful commercialization of PEMFC in automotive applications, rapid startup from subzero temperatures must be achieved, which is referred to as “cold start” of PEMFC. The major problem of PEMFC cold start is that the product water freezes when the temperature inside the PEMFC is lower than the freezing point of water. If the catalyst layer (CL) is fully covered by ice before the cell temperature rises above freezing point, the electrochemical reaction may be stopped due to the blockage of the reaction site. In addition, ice formation may also result in serious damage to the structure of the membrane electrode assembly (MEA). Therefore, for PEMFC in automotive applications, successful cold start is

of paramount importance. The U.S. Department of Energy (DOE) already established a challenging technical target: rapid startup of PEMFC to 90% rated power from -20°C for 30 s in 2010 [1]. At present, rapid unassisted cold start of PEMFC is still very difficult. Most of the present PEMFCs employ various assisted cold start methods such as: resistance heating using DC from batteries, coolant heating, hot air blowing, catalytic hydrogen/oxygen reaction inside the PEMFC, etc. United Technology Corporation (UTC) achieved 34% rated power in 30 s from -30°C , they also achieved unassisted cold start from -35°C , but the duration is not reported [2]. Ballard Mark1100 PEMFC power module also has cold start capability from -15°C [3]. Apparently PEMFC cold start capability still needs significant improvement, especially for unassisted cold start, this is because assisted cold start might increase the volume and weight of the system, as well as the operation complexity and installation costs. However, only few of the numerous previous numerical and experimental PEMFC studies paid attention to cold start, even more significant effort is needed in this area for PEMFC commercialization.

By comparing with the experimental studies focusing on normal operating conditions, the cold start related experimental studies are much less. Tajiri et al. [4,5] designed experimental procedures to mainly investigate the effects of MEA characteristics on PEMFC cold start, and the initial water content in Nafion membrane was controlled by different purging methods. They found that the best cold start capability is achieved with the lowest initial water content, because more product water can be taken by the membrane, result-

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ing in less water freezing in the CL. Hou et al. [6,7] investigated the performance degradations after freeze–thaw cycles, and they indicated that membrane degradation is not a major issue for PEMFC cold start. Cho et al. [8,9] studied the freeze–thaw cycles with a sub-freezing temperature of -10°C , they reported that the unpurged initial condition resulted in severe performance degradation, and the solution purge (purged with antifreezes such as methanol) might be an effective way to achieve fast startup. Similarly, other experimental studies that focused on the effects of purging methods [10], startup temperatures [11], and purging temperatures [12] on cold start performance were also carried out. Thompson et al. [13] experimentally determined the oxygen reduction reaction (ORR) kinetics for PEMFC operating at subzero temperatures, and no significant change in the ORR mechanism was found. Ge and Wang [14,15] conducted the first two in situ visualizations of ice formation on the cathode CL surface by cooling the transparent cells with coolant recirculation, and they have reported that the freezing point of water in the CL is around -1°C . Ishikawa et al. [16] also reported that the freezing temperature of liquid water in the CL changes between -0.1 and -2.2°C . All these studies [14–16] showed that liquid water freezes at the temperatures lower than 0°C , known as the freezing point depression (FPD). Sun et al. [17] described a new method called catalytic hydrogen/oxygen reaction assisted cold start by supplying mixed hydrogen and air (or oxygen) into the fuel cell, however, further investigations are needed for this method. McDonald et al. [18] conducted an ex situ freeze–thaw study on MEA, and they reported that no degradation was observed for dry MEA. Cappadonia et al. [19,20] conducted experimental measurements of Nafion membrane conductance at various subzero temperatures, and Thompson et al. [21] reported the conductivities of Nafion membrane at different subzero temperatures based on the measured membrane conductance and sample size. The measurements in [19–21] all observed sharp decrements of membrane conductance/conductivity at subzero temperatures. In [21], it was also found that such sharp decrements occurred when the water phase changes were observed, and the amount of non-frozen membrane water content at different subzero temperatures were also estimated.

Similar to the experimental studies, only a few numerical models for PEMFC cold start are available in literature. Sundaresan and Moore [22] conducted an analytical model for cold start of PEMFC stacks. This one-dimensional model could predict the temperature for each of the single cell by performing energy balance and heat transfer analysis, and this model could also reveal the effects of the endplate thermal mass and the internal heating method for PEMFC cold start. Khandelwal et al. [23] also conducted a one-dimensional thermal model, and similar to Sundaresan and Moore [22], the cold start capability of PEMFC stacks can be evaluated. They reported that adjusting the startup current density, coolant heating, isolation of stack endplates are all effective ways to optimize the cold start performance. Mao and Wang [24] developed an analytical model, not only for temperature, this one-dimensional model can also predict the amount of ice formation in CL, water transport, changes of cell voltage and current density, etc. Wang [25] also conducted an analytical study and defined some important parameters affecting PEMFC cold start performance. The analytical models in [22–25] can only roughly predict the PEMFC cold start performance, and in order to investigate the fundamental physics of PEMFC cold start, multi-dimensional and multiphase models are needed. However, very few literatures are related to this field [26–29]. Ahluwalia and Wang [26] conducted a simple two-dimensional cold start model for single PEMFCs, and they reported that high startup current density is favorable for rapid cold start, and they also investigated the effects of feed gas temperatures, operating pressure, and electrical heating on cold start performance. Mao et al. [27] developed a three-dimensional, multiphase model, Meng [28] conducted a two-

dimensional multiphase model for PEMFC cold start simulations, and based on the work in Mao et al. [27], Jiang et al. [29] further conducted non-isothermal cold start simulations for PEMFC. However, all the numerical models in [27–29] assumed instantaneous desublimation of water vapour to ice, and the reaction product is water vapour. Since no liquid water is considered, so that these models can only provide reasonable results when the temperature is lower than the freezing point of water.

Based on the literature review, it is found that development of multi-dimensional, multiphase PEMFC cold start model with the capability to consider the detailed phase changes of water in both the membrane electrolyte and the pore volumes of the GDL and CL needs to be carried out for better predictions of cold start per-

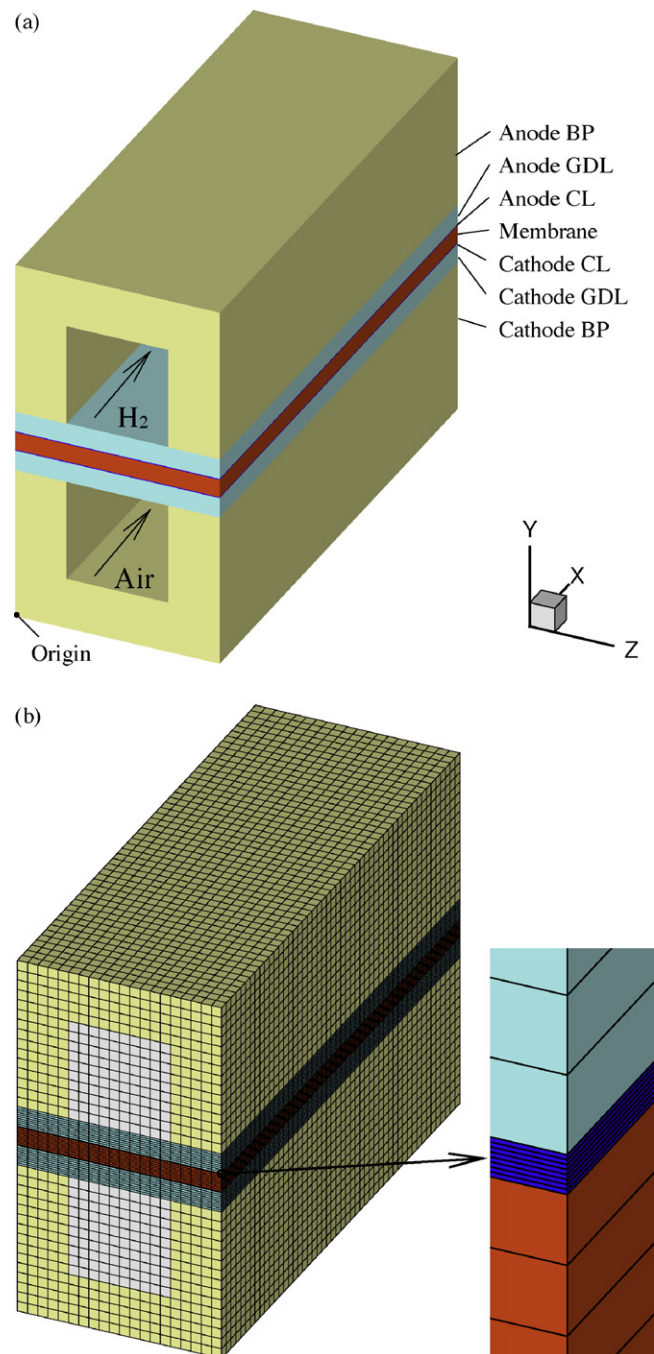


Fig. 1. Computational domain (a) and mesh (b) ($10\times$ minification along the x-direction).

formance. Therefore, in this study, a mathematical model has been developed to simulate the cold start processes in a PEMFC. Numerical simulations have been conducted for a single PEMFC starting at different subzero temperatures. In the following sections, a detailed description of the mathematical model is given first, followed by the results and discussion and the conclusion.

2. Cold start model

2.1. Conservation equations

Fig. 1 shows the computational domain and mesh for the cold start simulations, it can be noticed that all the major components of a PEMFC are considered in this model, which are the bipolar plate (BP), flow channel, GDL, CL and membrane. The cell properties and operating conditions are listed in Table 1. Fourteen conservation equations are developed and to be solved simultaneously in different domains, as listed in Table 2. These equations include

the conservations of mass, momentum (x , y and z velocities), gas species (hydrogen, oxygen and water vapour), liquid water, ice, non-frozen membrane water content, frozen membrane water content, electronic potential, ionic potential, and energy. The corresponding source terms of the 14 conservation equations are listed in Table 3.

The first two equations in Table 2 represent the mass and momentum conservations of the gas mixture, and the next equation describes the species transport of hydrogen, oxygen, and water vapour, respectively. Superficial velocity is used in this model to satisfy the mass conservation at the interfaces between GDL (porous) and flow channel. All the gas species are assumed to be ideal gas. It should be noticed that the effects of porous regions, liquid water and ice on gaseous transport are considered by modifying the gas phase permeability in the momentum equations and effective species diffusivities in the species transport equations. The gas phase and liquid phase permeabilities (K_g and K_{lg} , respectively, m^2) depend on the intrinsic permeabilities of the porous materials (K_0 , m^2 , listed in Table 1) and the local volume fraction of liquid water

Table 1
Cell properties and operating conditions.

Parameter	Value
Channel length; width; depth; rib width	50; 1.0; 1.0; 1.0 mm
Thicknesses of membrane (Nafion 117); CL; GDL (carbon paper)	0.178; 0.01; 0.2 mm
Membrane density; equivalent weight	$\rho_{mem} = 1980 \text{ kg m}^{-3}$; $EW = 1100 \text{ kg kmol}^{-1}$
Volume fraction of ionomer in CL, and the porosities of CL; GDL	$\omega = 0.2$; $\varepsilon_{CL,GDL} = 0.5$; 0.6
Contact angles and pore radii of CL; GDL	$\theta_{CL,GDL} = 100^\circ$; $r_{CL,GDL} = 1.2 \times 10^{-8}$; $3.89 \times 10^{-5} \text{ m}$
Intrinsic permeabilities of CL; GDL	$K_{CL,GDL}^0 = 6.2 \times 10^{-13}$; $6.2 \times 10^{-12} \text{ m}^2$
Densities of CL; GDL; BP	$\rho_{CL,GDL,BP} = 1000 \text{ kg m}^{-3}$
Specific heat capacities of membrane; CL; GDL; BP	$(C_p)_{mem,CL,GDL,BP} = 833$; 3300 ; 568 ; $1580 \text{ J kg}^{-1} \text{ K}^{-1}$
Thermal conductivities of membrane; CL; GDL; BP	$k_{mem,CL,GDL,BP} = 0.95$; 1.0 ; 1.0 ; $20 \text{ W m}^{-1} \text{ K}^{-1}$
Electrical conductivities of CL; GDL; BP	$\kappa_{CL,GDL,BP} = 300$; 300 ; $20,000 \text{ S m}^{-1}$
Cell voltage	$V_{cell} = 0.3 \text{ V}$
Stoichiometry ratios at 1.0 A cm^{-2}	$\xi_{a,c} = 2.0$
Relative humidities of inlet gases	$RH_{a,c}^{in} = 0$
Inlet gas temperatures	$T_{a,c}^{in} = T_0 = \text{volume averaged cell temperature or } T_{a,c}^{in} = T_{surr} = \text{surrounding temperature}$
Absolute total pressure at outlets	$p_{a,c}^{out} = 1 \text{ atm}$
Surrounding temperatures and heat transfer coefficient	$T_{surr} = -3$, -10°C ; $h = 55 \text{ W m}^{-2} \text{ K}^{-1}$

Table 2
Conservation equations.

Property	Conservation equation	Domains solved in
Mass	$\frac{\partial}{\partial t} (\varepsilon(1 - s_{lq} - s_{ice})\rho_g) + \nabla \cdot (\rho_g \vec{u}_g) = S_m$	Flow channel, GDL, CL
Momentum	$\frac{\partial}{\partial t} \left(\frac{\rho_g \vec{u}_g}{\varepsilon(1 - s_{lq} - s_{ice})} \right) + \nabla \cdot \left(\frac{\rho_g \vec{u}_g \vec{u}_g}{\varepsilon^2(1 - s_{lq} - s_{ice})^2} \right) = -\nabla p_g + \mu_g \nabla \cdot \left(\nabla \left(\frac{\vec{u}_g}{\varepsilon(1 - s_{lq} - s_{ice})} \right) + \nabla \left(\frac{\vec{u}_g^T}{\varepsilon(1 - s_{lq} - s_{ice})} \right) \right) - \frac{2}{3} \mu_g \nabla \cdot \left(\nabla \cdot \left(\frac{\vec{u}_g}{\varepsilon(1 - s_{lq} - s_{ice})} \right) \right) + S_u$	Flow channel, GDL, CL
Gas species i : hydrogen, oxygen, water vapour	$\frac{\partial}{\partial t} (\varepsilon(1 - s_{lq} - s_{ice})\rho_g Y_i) + \nabla \cdot (\rho_g \vec{u}_g Y_i) = \nabla \cdot (\rho_g D_i^{eff} \nabla Y_i) + S_i$	Flow channel, GDL, CL
Liquid water	$\frac{\partial (\varepsilon s_{lq} \rho_{lq})}{\partial t} + \nabla \cdot (\rho_{lq} \vec{u}_{lq}) = \nabla \cdot (\rho_{lq} D_{lq} \nabla s_{lq}) + S_{lq}$	GDL, CL
Ice	$\frac{\partial (\varepsilon s_{ice} \rho_{ice})}{\partial t} = S_{ice}$	GDL, CL
Non-frozen membrane water content	$\frac{\rho_{mem}}{EW} \frac{\partial (\omega \lambda_{nf})}{\partial t} = \frac{\rho_{mem}}{EW} \nabla \cdot (\omega^{1.5} D_{nmw} \nabla \lambda_{nf}) + S_{nmw}$	Membrane, CL
Frozen membrane water content	$\frac{\rho_{mem}}{EW} \frac{\partial (\omega \lambda_f)}{\partial t} = S_{fmw}$	Membrane
Electronic potential	$0 = \nabla \cdot (\kappa_{ele}^{eff} \nabla \phi_{ele}) + S_{ele}$	CL, GDL, BP
Ionic potential	$0 = \nabla \cdot (\kappa_{ion}^{eff} \nabla \phi_{ion}) + S_{ion}$	Membrane, CL
Energy	$\frac{\partial}{\partial t} ((\rho C_p)_{fl,sl}^{eff} T) + \nabla \cdot ((\rho C_p)_{fl}^{eff} \vec{u}_g T) = \nabla \cdot (k_{fl,sl}^{eff} \nabla T) + S_T$	All the domains

Table 3
Source terms.

Source term	Unit
$S_m = S_{H_2} + S_{O_2} + S_{vp}$	$\text{kg m}^{-3} \text{ s}^{-1}$
$S_u = \begin{cases} -\frac{\mu_g}{K_g} \bar{u}_g & (\text{in CL and GDL}) \\ 0 & (\text{in other zones}) \end{cases}$	$\text{kg m}^{-2} \text{ s}^{-2}$
$S_{H_2} = \begin{cases} -\frac{j_a}{2F} M_{H_2} & (\text{in anode CL}) \\ 0 & (\text{in other zones}) \end{cases}, \quad S_{O_2} = \begin{cases} -\frac{j_c}{4F} M_{O_2} & (\text{in cathode CL}) \\ 0 & (\text{in other zones}) \end{cases}$	$\text{kg m}^{-3} \text{ s}^{-1}$
$S_{vp} = \begin{cases} -S_{v-l} - S_{v-i} + S_{n-v} M_{H_2O} & (\text{in CL}) \\ -S_{v-l} - S_{v-i} & (\text{in other zones}) \end{cases}$	$\text{kg m}^{-3} \text{ s}^{-1}$
$S_{lq} = S_{v-l} - S_{l-i}, \quad S_{ice} = \begin{cases} S_{v-i} + S_{l-i} + S_{n-i} M_{H_2O} & (\text{in CL}) \\ S_{v-i} + S_{l-i} & (\text{in other zones}) \end{cases}$	$\text{kg m}^{-3} \text{ s}^{-1}$
$S_{nmw} = \begin{cases} -S_{n-f} & (\text{in membrane}) \\ \frac{j_c}{2F} - S_{n-v} - S_{n-i} + S_{EOD} & (\text{in cathode CL}), \quad S_{mw} = S_{n-f} \\ -S_{n-v} - S_{n-i} + S_{EOD} & (\text{in anode CL}) \end{cases}$	$\text{kmol m}^{-3} \text{ s}^{-1}$
$S_{ele} = \begin{cases} -j_a & (\text{in anode CL}) \\ j_c & (\text{in cathode CL}) \\ 0 & (\text{in other zones}) \end{cases}, \quad S_{ion} = \begin{cases} j_a & (\text{in anode CL}) \\ -j_c & (\text{in cathode CL}) \\ 0 & (\text{in other zones}) \end{cases}$	A m^{-3}
$S_T = \begin{cases} j_a \eta_{act} + \nabla \phi_{ele} ^2 \kappa_{ele}^{eff} + \nabla \phi_{ion} ^2 \kappa_{ion}^{eff} + S_{pc} & (\text{in anode CL}) \\ -\frac{j_c T \Delta S}{2F} + j_c \eta_{act} + \nabla \phi_{ele} ^2 \kappa_{ele}^{eff} + \nabla \phi_{ion} ^2 \kappa_{ion}^{eff} + S_{pc} & (\text{in cathode CL}) \\ \nabla \phi_{ele} ^2 \kappa_{ele}^{eff} + S_{pc} & (\text{in GDL}) \\ \nabla \phi_{ele} ^2 \kappa_{ele}^{eff} & (\text{in BP}) \\ \nabla \phi_{ion} ^2 \kappa_{ion}^{eff} + S_{pc} & (\text{in membrane}) \\ 0 & (\text{in other zones}) \end{cases}$	W m^{-3}

(usually with an exponent of 4.0) [30,31]. In this work, the formation of ice is considered as well, by assuming that the effects of formations of ice and liquid are similar on the permeabilities, therefore, the gas phase and liquid phase permeabilities can be calculated as

$$K_g = K_0(1 - s_{lq} - s_{ice})^{4.0} \quad (1)$$

$$K_{lq} = K_0 s_{lq}^{4.0} (1 - s_{ice})^{4.0} \quad (2)$$

where both the liquid and ice volume fractions (s_{lq} and s_{ice} , respectively) are considered. Eq. (1) depicts that if the local pore volume of the GDL or CL is full of liquid water and ice, the gas phase permeability will become zero, resulting in an infinite negative value for the momentum source term (Table 3), which will result in zero velocity. Similarly, Eq. (2) represents that if the local pore volume of the GDL or CL is full of ice, the liquid phase permeability will also become zero.

The gas mixture density (kg m^{-3}) is calculated based on the ideal gas law:

$$\rho_g = p_g \left(RT \sum_i \frac{Y_i}{M_i} \right)^{-1} \quad (3)$$

where p_g is the gas phase pressure (Pa), T is the temperature (K), R is the universal gas constant ($\text{J K}^{-1} \text{ kmol}^{-1}$), and Y_i and M_i (kg kmol^{-1}) are the mass fraction and molecular weight of species i . The dynamic viscosity of the ideal gas mixture (μ_g , $\text{kg m}^{-1} \text{ s}^{-1}$) is calculated based on kinetic theory [32] as

$$\mu_g = \sum_i \frac{X_i \mu_i}{\sum_j X_j \psi_{ij}} \quad (4)$$

$$\psi_{ij} = \frac{[1 + (\mu_i/\mu_j)^{0.5} (M_i/M_j)^{0.25}]^{0.5}}{[8(1 + M_i/M_j)]^{0.5}} \quad (5)$$

and X_i is the mole fraction of species i , and i and j represent different species. Since the present cold start model provides the capability to simulate PEMFC operation from subzero to normal operating temperatures, therefore, the changes of different parameters with temperature must be considered, and Table 4 shows the correlations of the dynamic viscosity and mass diffusivity of each species with the changes of temperature and pressure based on the correlations and experimental data in [32]. In GDL and CL, the Bruggemann correlation is used to consider the effects of porosity and tortuosity on the effective gaseous diffusivities, and in some multiphase models (e.g. in [33]) the effect of liquid water on the effective gaseous diffusivities is treated the same as porosity with an exponent of 1.5. In this model, the effect of ice formation is treated the same as liquid water, and the diffusivity of each species is corrected by porosity (ϵ , listed in Table 1) and volume fractions of liquid water and ice:

$$D_i^{eff} = D_i \epsilon^{1.5} (1 - s_{lq} - s_{ice})^{1.5} \quad (6)$$

where D_i ($\text{m}^2 \text{ s}^{-1}$) is the mass diffusivity for species i calculated based on the correlations in Table 4. Apparently, if the local pore volume of the GDL and CL is full of liquid water and ice, the diffusivity of each species will become zero, reflecting the blockage due to liquid water and ice.

The reaction rates (j_a and j_c , A m^{-3}) for both the anode and cathode CLs are calculated by using the Butler–Volmer equation:

$$j_a = (1 - s_{lq} - s_{ice}) j_{0,a}^{ref} \left(\frac{c_{H_2}}{c_{H_2}^{ref}} \right)^{0.5} \times \left[\exp \left(\frac{2\alpha_a F}{RT} \eta_{act} \right) - \exp \left(-\frac{2\alpha_c F}{RT} \eta_{act} \right) \right] \quad (7)$$

Table 4
Transport parameters.

Parameter	Correlation/value (T in K, p in Pa)	Unit
Hydrogen dynamic viscosity	$\mu_{H_2} = 3.205 \times 10^{-3} (T/293.85)^{1.5} (T + 72)^{-1.0}$	$\text{kg m}^{-1} \text{s}^{-1}$
Oxygen dynamic viscosity	$\mu_{O_2} = 8.46 \times 10^{-3} (T/292.25)^{1.5} (T + 127)^{-1.0}$	$\text{kg m}^{-1} \text{s}^{-1}$
Water vapour dynamic viscosity	$\mu_{vp} = 7.512 \times 10^{-3} (T/291.15)^{1.5} (T + 120)^{-1.0}$	$\text{kg m}^{-1} \text{s}^{-1}$
Hydrogen and water vapour diffusivities in anode	$D_{H_2}^a = D_{vp}^a = 1.055 \times 10^{-4} (T/333.15)^{1.5} (101325/p)$	$\text{m}^2 \text{s}^{-1}$
Oxygen diffusivity in cathode	$D_{O_2}^c = 2.652 \times 10^{-5} (T/333.15)^{1.5} (101325/p)$	$\text{m}^2 \text{s}^{-1}$
Water vapour diffusivity in cathode	$D_{vp}^c = 2.982 \times 10^{-5} (T/333.15)^{1.5} (101325/p)$	$\text{m}^2 \text{s}^{-1}$
Liquid water dynamic viscosity	$\mu_{lq} = 2.414 \times 10^{-5} \times 10^{247.8/(T-140)}$	$\text{kg m}^{-1} \text{s}^{-1}$
Liquid water and ice densities	$\rho_{lq} = 990, \rho_{ice} = 920$	kg m^{-3}
Evaporation, condensation, desublimation, fusion, melting rates	$\gamma_{evap,cond,desb,fusn,melt} = 1.0$	s^{-1}
Water transfer rates	$\zeta_{n-v,v-n,n-i,n-f,f-n} = 1.0$	s^{-1}
Latent heat of fusion	$h_{fusn} = 3.336 \times 10^5$	J kg^{-1}
Specific heat capacities of different gas species, liquid water, and ice	$(C_p)_{H_2} = 14, 283, (C_p)_{O_2} = 919.31, (C_p)_{vp} = 2014, (C_p)_{lq} = 4182, (C_p)_{ice} = 2050$	$\text{J kg}^{-1} \text{K}^{-1}$
Thermal conductivities of different gas species, liquid water, and ice	$k_{H_2} = 0.1672, k_{O_2} = 0.0246, k_{vp} = 0.0261, k_{lq} = 0.6, k_{ice} = 2.3$	$\text{W m}^{-1} \text{K}^{-1}$
Entropy change of reaction	$\Delta S = -163, 110$	$\text{J kmol}^{-1} \text{K}^{-1}$

$$j_c = (1 - s_{lq} - s_{ice}) j_{0,c}^{ref} \frac{C_{O_2}}{C_{O_2}^{ref}} \left[-\exp\left(\frac{4\alpha_a F}{RT} \eta_{act}\right) + \exp\left(-\frac{4\alpha_c F}{RT} \eta_{act}\right) \right] \quad (8)$$

where F is Faraday's constant (C mol^{-1}), and a linear relationship to reflect the blockage of the reaction sites due to the ice and liquid water is assumed [27–29]. The correlations and values of the parameters related to electrochemical reactions are listed in Table 5. It should be noticed that the reference exchange current densities also vary with temperature due to the change of electrochemical reaction kinetics [30], and the correlations are given in this table as well.

Based on the densities of liquid water and ice tabulated in [34], it is found that the changes of liquid water and ice densities with temperature are not very significant, therefore, in this model, liquid water and ice are assumed to have constant densities, and the values are listed in Table 4. The interfacial drag coefficient (ι) in liquid water conservation equation is defined [35] as

$$\iota = \frac{K_{lq} \mu_g}{K_g \mu_{lq}} \quad (9)$$

Since the liquid water dynamic viscosity (μ_{lq}) strongly depends on temperature, therefore, a correlation for temperature dependence of liquid water viscosity [36] is used rather than using a constant value, as listed in Table 4. The diffusivity of liquid water (D_{lq} , $\text{m}^2 \text{s}^{-1}$) in GDL and CL is defined [35] by considering the effects of capillary pressure:

$$D_{lq} = -\frac{K_{lq}}{\mu_{lq}} \frac{dp_c}{ds_{lq}} \quad (10)$$

and the capillary pressure (p_c , Pa) is calculated as a function of liquid water volume fraction (the Leverett function) [37,38] as

$$p_c = \sigma \cos \theta \left(\frac{\varepsilon}{K_0} \right)^{0.5} \times [1.42(1-s_{lq}) - 2.12(1-s_{lq})^2 + 1.26(1-s_{lq})^3] \quad \theta < 90^\circ$$

$$p_c = \sigma \cos \theta \left(\frac{\varepsilon}{K_0} \right)^{0.5} \times [1.42s_{lq} - 2.12s_{lq}^2 + 1.26s_{lq}^3] \quad \theta > 90^\circ \quad (11)$$

where θ is the contact angle (listed in Table 1), which depends on the wettabilities of GDL and CL; σ (N m^{-1}) is the surface tension between liquid water and gas, based on the experimental data tabulated in [34], the following linear correlation is derived to account for the temperature dependence between 273.15 and 373.15 K:

$$\sigma = -0.0001676T + 0.1218; \quad 273.15 \text{ K} \leq T \leq 373.15 \text{ K} \quad (12)$$

where the units of σ and T are N m^{-1} and K, respectively. The liquid phase pressure (p_{lq} , Pa) is calculated based on the capillary pressure (p_c , Pa) and gas phase pressure (p_g , Pa) [39] as

$$p_{lq} = p_g - p_c \quad (13)$$

Table 5
Parameters related to electrochemical reactions.

Parameter	Correlation/value (T in K)	Unit
Overpotential (activation loss)	$\eta_{act} = \phi_{ele} - \phi_{ion}$	V
Transfer coefficient	$\alpha_a = \alpha_c = 0.5$	Dimensionless
Volumetric reference exchange current density in anode	$j_{0,a}^{ref} = j_{0,a}^{ref} _{353.15 \text{ K}} \exp\left[-1400\left(\frac{1}{T} - \frac{1}{353.15}\right)\right], \quad j_{0,a}^{ref} _{353.15 \text{ K}} = 10^9$	A m^{-3}
Volumetric reference exchange current density in cathode	$j_{0,c}^{ref} = j_{0,c}^{ref} _{353.15 \text{ K}} \exp\left[-7900\left(\frac{1}{T} - \frac{1}{353.15}\right)\right], \quad j_{0,c}^{ref} _{353.15 \text{ K}} = 10^4$	A m^{-3}
Reference hydrogen, oxygen concentrations	$C_{H_2}^{ref} = C_{O_2}^{ref} = 40.0$	mol m^{-3}

and the liquid phase velocity (m s^{-1}) can be calculated based on the liquid phase pressure [40] as

$$\bar{u}_{lq} = -\frac{K_{lq}}{\mu_{lq}} \nabla p_{lq} \quad (14)$$

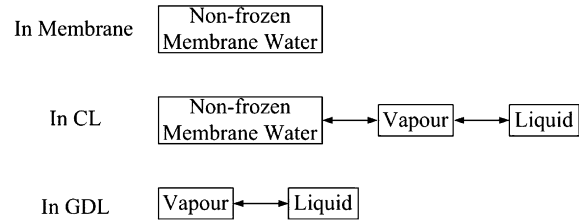
Alternatively, other capillary functions can also be used in this model, such as the one used in [30] (p_c in Pa)

$$\begin{aligned} p_c &= 35.6 - 2.09 \exp(44.9s_{lq} - 14.41) \\ &\quad + 2.09 \exp(-22.2s_{lq} + 7.13) \quad (\text{in GDL}) \\ p_c &= -2395 - 2431 \exp(92.36s_{lq} - 52.37) \\ &\quad + 2431 \exp(-0.0088s_{lq} + 0.005) \quad (\text{in CL}) \end{aligned} \quad (15)$$

The numerical simulations in [41] compared the effects of the different capillary functions on liquid water volume fraction, and it was noticed that even the liquid water volume fractions are slightly different with different capillary functions, the general liquid water distributions are similar. Therefore, in this study, the Leverett function (Eq. (11)) is used because of its numerical stability. In addition, liquid water is only formed during a successful cold start process (ice melts and saturated vapour condenses to liquid water, also explained later on in this paper), therefore, the effects of the different capillary functions on the ice melting and vapour condensing processes will be investigated in the future studies.

Fig. 2 presents the schematics of water phase change and transfer for both the normal operating condition and cold start. It should be noticed that the schematics in Fig. 2 are for the operating conditions that partially or non-humidified gases are supplied, and in this case liquid water is formed only if the gas stream is fully humidified first (enough amount of water taken by the gas stream from the membrane electrolyte to reach saturation). Based on this premise, the membrane hydration/dehydration is taken into account by considering the mass transfer between vapour and non-frozen membrane water in CL. On the other hand, if fully humidified gases are supplied, the water leaving the membrane electrolyte will become liquid water directly (water vapour is already saturated), and therefore the mass transfer between liquid and non-frozen membrane water needs to be considered in this case. More discussions on the membrane hydration/dehydration processes can be found in [41]. For PEMFC cold start, partially or non-humidified gases are usually supplied to reduce the amount of ice formation [4,5,27,29]. Therefore, in this study, only non-humidified inlet gases are considered (listed in Table 1), and hence, the membrane hydration/dehydration is taken into account by considering the mass

Normal Operating Condition:



Cold Start:

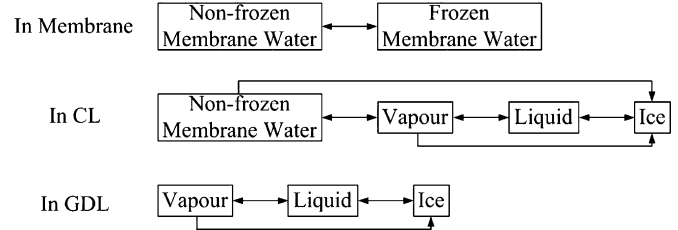


Fig. 2. Schematics of water phase change and transfer (with partially or non-humidified gases supplied).

transfer between vapour and non-frozen membrane water in CL (shown in Fig. 2). By comparing with the normal operating condition, frozen water in membrane and ice in CL and GDL are further included for cold start.

Fusion and melting of water in the membrane electrolyte are considered based on the experimental investigation in [21], which also found that the frozen membrane water does not contribute to the ion transport. In membrane, fusion and melting between frozen and non-frozen membrane water is considered. In CL, for simplicity, it is assumed that the non-frozen membrane water freezes to ice, and ice melts to liquid water, based on the assumption that the frozen water does not contribute to the ion transport [21]. The phase changes between vapour, liquid water and ice are fully considered in CL and GDL. It should be noticed that because the operating pressure of PEMFC is always equal to or greater than the atmospheric pressure, therefore the sublimation process (phase change from ice directly to vapour) can be safely neglected.

Table 6 shows the phase change functions in considering the phase changes between vapour, liquid water and ice in GDL and CL, and Table 7 shows the phase change and water transfer functions related to membrane water content. The values of phase change rates and water transfer rates are listed in Table 4 [30,35,41–43]. p_{sat} (Pa) is the saturation pressure of vapour, and by checking the

Table 6
Water phase change functions.

Phase change function	Unit
Vapour and liquid water	
$S_{v-l} = \begin{cases} \gamma_{\text{cond}} \epsilon (1 - s_{lq} - s_{ice}) \frac{(p_g X_{vp} - p_{\text{sat}}) M_{\text{H}_2\text{O}}}{RT} & \text{if } p_g X_{vp} \geq p_{\text{sat}} \\ \gamma_{\text{evap}} \epsilon s_{lq} \frac{(p_g X_{vp} - p_{\text{sat}}) M_{\text{H}_2\text{O}}}{RT} & \text{if } p_g X_{vp} < p_{\text{sat}} \\ 0 & \text{if } T < T_N + T_{FPD} \end{cases}$	$\text{kg m}^{-3} \text{ s}^{-1}$
Vapour and ice	
$S_{v-i} = \begin{cases} \gamma_{\text{desb}} \epsilon (1 - s_{lq} - s_{ice}) \frac{(p_g X_{vp} - p_{\text{sat}}) M_{\text{H}_2\text{O}}}{RT} & \text{if } p_g X_{vp} \geq p_{\text{sat}} \\ 0 & \text{if } p_g X_{vp} < p_{\text{sat}} \\ 0 & \text{if } T \geq T_N + T_{FPD} \end{cases}$	$\text{kg m}^{-3} \text{ s}^{-1}$
Liquid water and ice	
$S_{l-i} = \begin{cases} \gamma_{\text{fusn}} \epsilon s_{lq} \rho_{lq} & \text{if } T < T_N + T_{FPD} \\ -\gamma_{\text{melt}} \epsilon s_{ice} \rho_{ice} & \text{if } T \geq T_N + T_{FPD} \end{cases}$	$\text{kg m}^{-3} \text{ s}^{-1}$

Table 7
Membrane water related phase change and water transfer functions.

Phase change/water transfer function	Unit
Non-frozen membrane water and vapour $S_{n-v} = \zeta_{n-v} \frac{\rho_{mem}}{EW} (\lambda_{nf} - \lambda_{equil})(1 - S_{lq} - S_{ice})$	$\text{kmol m}^{-3} \text{ s}^{-1}$
Non-frozen membrane water and ice $S_{n-i} = \begin{cases} \zeta_{n-i} \frac{\rho_{mem}}{EW} (\lambda_{nf} - \lambda_{sat})(1 - S_{lq} - S_{ice}) & \text{if } \lambda_{nf} \geq \lambda_{sat} \\ 0 & \text{if } \lambda_{nf} < \lambda_{sat} \end{cases}$	$\text{kmol m}^{-3} \text{ s}^{-1}$
Non-frozen membrane water and frozen membrane water $S_{n-f} = \begin{cases} \zeta_{n-f} \frac{\rho_{mem}}{EW} (\lambda_{nf} - \lambda_{sat}) & \text{if } \lambda_{nf} \geq \lambda_{sat} \\ \zeta_{n-f} \frac{\rho_{mem}}{EW} \lambda_f & \text{if } \lambda_{nf} < \lambda_{sat} \end{cases}$	$\text{kmol m}^{-3} \text{ s}^{-1}$
Membrane water transfer due to EOD $S_{EOD} = \nabla \cdot \left(\frac{n_d}{F} \kappa_{ion}^{eff} \nabla \phi_{ion} \right)$	$\text{kmol m}^{-3} \text{ s}^{-1}$

experimental data tabulated in [34], it is found that the correlation provided by Springer et al. [44] provides acceptable agreement with the experimental data in the temperature range from -50 to 100°C , therefore can be used in this model:

$$\log_{10} \left(\frac{p_{sat}}{101,325} \right) = -2.1794 + 0.02953(T - 273.15) - 9.1837 \times 10^{-5}(T - 273.15)^2 + 1.4454 \times 10^{-7}(T - 273.15)^3 \quad (16)$$

where the units of p_{sat} and T are Pa and K, respectively. The experiments in [14–16] all observed that the fusion and desublimation of water in CL take place at the temperatures below the normal freezing point of water (273.15 K), which is mainly due to the wettability and extremely small pore size in CL, known as the Gibbs–Thomson undercooling [45]. The difference between the freezing point in small pores and normal freezing point of water ($T_N = 273.15$ K) is defined as T_{FPD} (K), and can be calculated as

$$T_{FPD} = \frac{T_N \sigma_{|273.15 \text{ K}} \cos \theta}{\rho_{ice} h_{fusin} r_{CL, GDL}} \quad (17)$$

where h_{fusin} (J kg^{-1}) is the latent heat of fusion as listed in Table 4, and $r_{CL, GDL}$ (m) is the pore radius of CL or GDL, as listed in Table 1. The units for σ and ρ_{ice} are N m^{-1} and kg m^{-3} , respectively. Based on the values listed in Tables 1 and 4, T_{FPD} is calculated to be about -1 K in CL by using Eq. (17), which agrees with the experimental investigations in [14,15].

Since the three-phase contact in CL must be satisfied for PEMFC operation, which requires certain amount of ionomer in CL, resulting in certain amount of membrane water content. Therefore, in this model, the transport of non-frozen membrane water content (λ_{nf}) is also solved in CL. The product water is assumed to be non-frozen membrane water because the electrochemical reactions only take place at the ionomer/catalyst interface; similar assumption was also used in [30]. In the conservation equations of non-frozen and frozen membrane water contents in Table 2, ρ_{mem} and EW represent the dry membrane density (kg m^{-3}) and equivalent weight of membrane (kg kmol^{-1}), respectively (listed in Table 1). The relationship between membrane water content and mole concentration of water in membrane (c , kmol m^{-3}) is defined as

$$\lambda_{nf, f} = \frac{EW}{\rho_{mem}} c_{nf, f} \quad (18)$$

The diffusivity of non-frozen membrane water content (D_{nmw} , $\text{m}^2 \text{ s}^{-1}$) can be calculated [46] as

$$D_{nmw} = \begin{cases} 3.1 \times 10^{-7} \lambda_{nf} [\exp(0.28 \lambda_{nf}) - 1] \exp \left(\frac{-2346}{T} \right) & (\text{if } 0 < \lambda_{nf} < 3) \\ 4.17 \times 10^{-8} \lambda_{nf} [161 \exp(-\lambda_{nf}) + 1] \exp \left(\frac{-2346}{T} \right) & (\text{if } 3 \leq \lambda_{nf} < 17) \end{cases} \quad (19)$$

where the units of D_{nmw} and T are $\text{m}^2 \text{ s}^{-1}$ and K, respectively. The equilibrium membrane water content (water uptake), λ_{equil} , is calculated by using the correlation in [44]

$$\lambda_{equil} = \begin{cases} 0.043 + 17.81a - 39.85a^2 + 36.0a^3 & \text{if } 0 \leq a \leq 1 \\ 14.0 + 1.4(a - 1) & \text{if } 1 < a \leq 3 \end{cases} \quad (20)$$

where a is the water activity, defined as

$$a = \frac{X_{vp} p_g}{p_{sat}} + 2S_{lq} \quad (21)$$

It should be noticed that Eq. (20) is based on experimental measurements at 30°C . However, this is probably the most suitable correlation for this cold start model, because most of the other correlations are based on experimental measurements at 80°C . Eq. (20) was also used in other numerical models for cold start simulations [27,29]. The difference between λ_{equil} and λ_{nf} indicates whether water transfers into or out of membrane, and as mentioned earlier, the mass transfer between vapour and non-frozen membrane water is considered during this process. At subzero temperatures, the experimental study in [21] observed that the maximum allowed amount of non-frozen membrane water decreases with the decrement of temperature, and based on the experimental results shown in [21], the following correlation is fitted to calculate the maximum allowed non-frozen membrane water content at different subzero temperatures (T in K), namely, saturated membrane water content (λ_{sat})

$$\lambda_{sat} = \begin{cases} 4.837 & (\text{if } T < 223.15 \text{ K}) \\ [-1.304 + 0.01479T - 3.594 \times 10^{-5} T^2]^{-1} & (\text{if } 223.15 \text{ K} \leq T < T_N) \\ > \lambda_{nf} & (\text{if } T \geq T_N) \end{cases} \quad (22)$$

It should be noticed that since no further water phase change was detected when the temperature is lower than -50°C [21], therefore the saturated membrane water content remains constant in this temperature range. Moreover, the membrane water does not freeze if the temperature is higher than the normal freezing temperature of water ($T_N = 273.15$ K), therefore the saturated membrane water content is always higher than the non-frozen membrane water content in this temperature range. The effects on water transfer due to the electro-osmotic drag (EOD) is considered by adding a water transfer function (S_{EOD}) into the source term of the non-frozen membrane water conservation equation, as shown in Table 7. The EOD drag coefficient (n_d) is calculated [44] as

$$n_d = \frac{2.5 \lambda_{nf}}{22} \quad (23)$$

The EOD coefficient is the number of water molecules dragged per hydrogen ion moved through the ionomer.

The transient terms in the conservation equations of the electronic and ionic potentials listed in Table 2 are neglected, due to fast electrochemical double layer changing and discharging process [47]. The ion conductivity of the ionomer (κ_{ion} , S m^{-1}) is given [44] as

$$\kappa_{ion} = (0.5139 \lambda_{nf} - 0.326) \exp \left[1268 \left(\frac{1}{303.15} - \frac{1}{T} \right) \right] \quad (24)$$

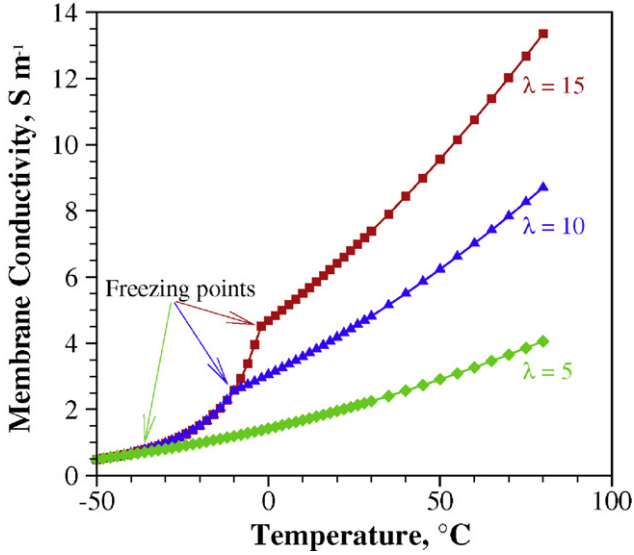


Fig. 3. Effects of water freezing on the membrane conductivity.

where the units of κ_{ion} and T are $S m^{-1}$ and K, respectively. It should be noticed that Eq. (24) is correlated based on experimental measurements between 30 and 80 °C, and it overestimates the ion conductivity at subzero temperatures (sharper decrements of the conductivity were observed [19–21]). However, by considering the freezing of membrane water at subzero temperatures, Eq. (24) with non-frozen membrane water content rather than total membrane water content provides a more reasonable agreement with the experimental measurements in [21]. Fig. 3 shows the changes of ion conductivity at different total membrane water contents, it can be observed that the ion conductivity starts dropping fast when the membrane water starts freezing. The effective electron conductivity in CL and GDL and ion conductivity in CL are further modified with the porosities (ε) of the CL and GDL and the ionomer volume fraction in CL (ω). Based on the Bruggemann correlation, an exponent of 1.5 is used:

$$\kappa_{ele}^{eff} = (1 - \varepsilon - \omega)^{1.5} \kappa_{ele} \quad (25)$$

$$\kappa_{ion}^{eff} = \omega^{1.5} \kappa_{ion} \quad (26)$$

The effective volumetric heat capacities ($J m^{-3} K^{-1}$) in the energy conservation equation in Table 2 are determined as

$$(\rho C_p)_{fl,sl}^{eff} = \varepsilon [s_{lq} \rho_{lq} (C_p)_{lq} + s_{ice} \rho_{ice} (C_p)_{ice} + (1 - s_{lq} - s_{ice}) \rho_g (C_p)_g] + (1 - \varepsilon - \omega) \rho_{sld} (C_p)_{sld} + \omega \rho_{mem} (C_p)_{mem} \quad (27)$$

$$(\rho C_p)_{fl}^{eff} = \varepsilon [s_{lq} \rho_{lq} (C_p)_{lq} + (1 - s_{lq} - s_{ice}) \rho_g (C_p)_g] \quad (28)$$

where $(C_p)_g$ ($J kg^{-1} K^{-1}$) is the specific heat capacity of the gas mixture, and it is determined based on the mass fraction of each species:

$$C_p = \sum_i Y_i (C_p)_i \quad (29)$$

where $(C_p)_{sld}$ ($J kg^{-1} K^{-1}$) represents the specific heats of the electron conducting materials (platinum powders, carbon powders, etc.) in CL and all the solid materials in GDL and BP. For simplicity, the effective thermal conductivity ($W m^{-1} K^{-1}$) in the energy conservation equation is assumed to be a volume averaged value:

$$k_{fl,sl}^{eff} = \varepsilon [s_{lq} k_{lq} + s_{ice} k_{ice} + (1 - s_{lq} - s_{ice}) k_g] + (1 - \varepsilon - \omega) k_{sld} + \omega k_{mem} \quad (30)$$

where the thermal conductivity of the gas mixture, k_g ($W m^{-1} K^{-1}$), is calculated based on kinetic theory [32] as

$$k_g = \sum_i \frac{X_i k_i}{\sum_j X_j \psi_{ij}} \quad (31)$$

$$\psi_{ij} = \frac{[1 + (k_i/k_j)^{0.5} (M_i/M_j)^{0.25}]^{0.5}}{[8(1 + M_i/M_j)]^{0.5}} \quad (32)$$

The specific heat capacities and thermal conductivities are all assumed to be constants for all the substances in this model. Based on the values tabulated in [34], it is found that the changes of these two parameters of different substances between –50 and 100 °C are not very significant, therefore this assumption is acceptable for the present model. The values of these two parameters for different cell components, gas species, liquid water and ice are all given in Tables 1 and 4. The latent heat function, S_{pc} ($W m^{-3}$), is defined based on the water phase change and transfer rates:

$$S_{pc} = \begin{cases} h_{cond} S_{v-l} + (h_{cond} + h_{fussn}) S_{v-i} + h_{fussn} S_{l-i} & (\text{in GDL}) \\ h_{cond} (S_{v-l} - S_{n-v} M_{H_2O}) + (h_{cond} + h_{fussn}) S_{v-i} + h_{fussn} (S_{l-i} + S_{n-i} M_{H_2O}) & (\text{in CL}) \\ h_{fussn} S_{n-f} M_{H_2O} & (\text{in membrane}) \\ 0 & (\text{in other zones}) \end{cases} \quad (33)$$

It should be noticed that the non-frozen membrane water is assumed to be in liquid phase, this assumption is also used in the experimental study in [21]. The water vapour absorption into a membrane is a process similar to a vapour condensation process and involves a release of heat, similarly, water molecules leaving a membrane as vapour is similar to a liquid evaporation process and requires heat; by comparing these processes, Ostrovskii and Gostev [48] showed that the difference in the amounts of heat released or absorbed is very small, and the numerical simulations in [41] also confirmed that such difference can be safely neglected. Moreover, the specific or molar entropy of the non-frozen membrane water is also not available in literature. Therefore, it is reasonable to assume that the non-frozen membrane water is the same as liquid water. Based on this assumption, the product water is also in liquid phase, and the entropy change of the overall reaction, ΔS ($J kmol^{-1} K^{-1}$), is defined based on liquid water product, as listed in Table 4. A constant value of ΔS is used by neglecting the temperature effects (based on the values tabulated in [34]). The temperature dependence of the latent heat of condensation, h_{cond} ($J kg^{-1}$), is considered by fitting the values tabulated in [34]

$$h_{cond} = -2438.5T + 3,170,700 \quad (34)$$

where the units of h_{cond} and T are $J kg^{-1}$ and K, respectively.

2.2. Boundary conditions

The domains that different conservation equations are solved in are listed in Table 2. The mass, momentum and gas species conservation equations are solved in flow channel, GDL and CL. Liquid water, ice and non-frozen membrane water conservation equations are solved in CL, and both the non-frozen and frozen membrane water conservation equations are solved in membrane, therefore the detailed phase changes between the five phases of water are fully accounted for. The electronic potential and ionic potential conservation equations are all solved in CL, therefore the activation loss due to the electrochemical double layer charging and discharging can be considered. Energy conservation equation is solved in all the domains to fully account for all the heat transfer and generation processes.

At the inlets of the flow channels, the mass flow rates (kg s^{-1}) are defined as

$$\dot{m}_a = \frac{\rho_g^a \xi_a I_{ref} A}{2F c_{H_2}}, \quad \dot{m}_c = \frac{\rho_g^c \xi_c I_{ref} A}{4F c_{O_2}} \quad (35)$$

$$c_{H_2} = \frac{(p_g^a - RH_a p_{sat})}{RT_{in}^a}, \quad c_{O_2} = \frac{0.21(p_g^c - RH_c p_{sat})}{RT_{in}^c} \quad (36)$$

where ξ_a and ξ_c are the stoichiometry ratios (listed in Table 1) for the anode and cathode, respectively. I_{ref} is the reference current density (A cm^{-2} , listed in Table 1), A is the flat active surface area of the CL (m^2). Since the cold start model only considers part of the PEMFC (part of a single cell with straight channel), the inlet gases are assumed to be already heated up by the cell. Therefore, constant temperatures are defined at the anode and cathode inlets (T_{in} , K) to be equal to the volume averaged cell temperature (T_0 , K) for each time step. If the straight cell for the simulations is at the inlet of the whole cell, the inlet gas temperatures should be equal to the ambient temperature. Therefore, numerical simulations are carried out to compare these two inlet gas conditions (the inlet gas temperatures equal to the ambient temperatures or the cell temperatures, respectively), and no noticeable difference is found for these two conditions, as shown in Section 3.2. Furthermore, constant pressures are defined at the flow channel outlets, as listed in Table 1.

The liquid water volume fraction is fixed at zero in the flow channels, reflecting the condition that liquid water can be removed from flow channel quickly. However, for large-scale simulations with very long flow channels, liquid water may accumulate in the flow channel, and liquid water transport in flow channel also needs to be considered. Therefore, the cold start model will be further modified for large-scale simulations in the future.

The electronic potentials on the end surfaces of the BPs for both the anode and cathode are defined as

$$\phi_{ele}^{c,end} = 0; \quad \phi_{ele}^{a,end} = V_{rev} - V_{cell} = \eta_{total} \quad (37)$$

where V_{rev} and V_{cell} are the reversible and operating voltages, therefore η_{total} represents the total voltage loss. For liquid water product, the reversible cell voltage is calculated as

$$V_{rev} = 1.23 - 0.9 \times 10^{-3}(T_0 - 298) + \frac{RT_0}{2F} \left(\ln p_{H_2}^{in} + \frac{1}{2} \ln p_{O_2}^{in} \right) \quad (38)$$

In this equation, the inlet partial pressures of hydrogen and oxygen (atm), and the operating temperature (volume average temperature) (T_0 , K) are used for the calculation.

On the surrounding walls of the cell, a heat transfer coefficient is defined based on normal atmosphere condition [29,49]. Therefore, the heat transfer rate between the cell wall and surroundings can be calculated as

$$\dot{Q} = hA_{wall}(T_{surr} - T_{wall}) \quad (39)$$

where $\dot{Q}$ (W) is the heat transfer rate between the cell wall and surroundings, a positive value indicates that heat is absorbed by the cell, and vice versa; h is the heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$); A_{wall} is the surface area of the cell wall (m^2); T_{surr} is the temperature of the surrounding (K); and T_{wall} is the temperature of the cell wall (K).

2.3. Numerical procedures

The governing equations are discretized and solved in the commercial CFD software package Fluent 6.3 based on finite volume method. Equations and physical properties are written and implemented by using its user-defined function (UDF). Because cold start

simulations are time dependent, the PISO algorithm is used which is based on predictor–corrector splitting for unsteady problems. An algebraic multigrid (AMG) method with a Gauss–Seidel type smoother is used to accelerate the convergence. Adaptive time stepping is employed, and for the simulations presented in this paper, a minimum of 10^{-6} s and a maximum of 0.1 s time step sizes are used. Strict convergence criteria with a residual of 10^{-8} are used for all variables. The total number of cell shown in Fig. 1 is 76,000, and structured mesh is used for the whole computational domain. Each cell in the flow channel and BP has the same size with dimensions of $1.0 \text{ mm} \times 0.1 \text{ mm} \times 0.1 \text{ mm}$, along the x -, y - and z -directions, respectively. The mesh size along the y -direction is reduced to 0.0178, 0.001 and 0.02 mm for the membrane, CL and GDL, respectively, to ensure at least 10 cells along the y -direction in each layer. Grid independent study was carefully carried out. The difference in results (variations of cell temperature, current density, water content, ice formation) by further increasing the number of cells (up to two times of the one used) is negligible. Such meshing systems were also considered sufficient and widely used in previous three-dimensional multiphase simulations (for both the subzero and normal operating temperatures) [27,29,30].

3. Results and discussion

3.1. Comparison between model predictions and experimental data

In order to examine the accuracy of the present cold start model, the evolutions of the cell voltages for both the successful (from -3°C) and failed (from -20°C) cold start processes obtained from the experimental measurements in [5] are compared with the model predictions, as shown in Fig. 4. For the results shown in Fig. 4, some of the operating conditions and cell properties are changed to be consistent with the experimental conditions in [5] rather than using the values in Table 1. The startup current density is 0.04 A cm^{-2} for both of the cold start processes from -3 and -20°C , respectively; the stoichiometry ratio is 2.0 corresponding to the operating current density for both the anode and cathode

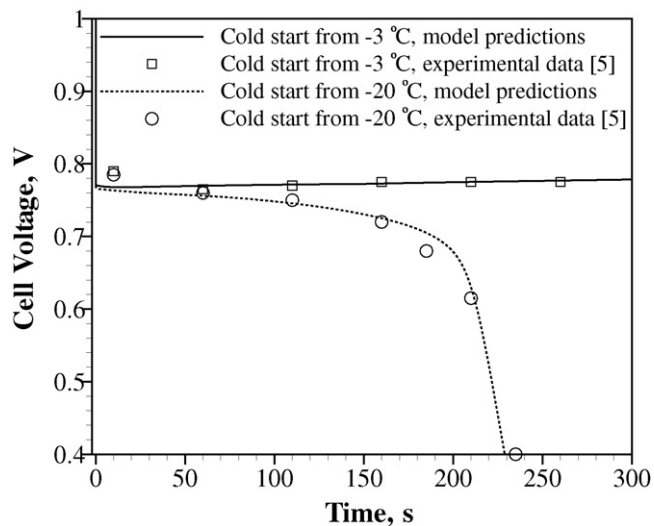


Fig. 4. Comparison between the present model predictions and experimental data [5] under the conditions: the current density is 0.04 A cm^{-2} for both of the cold start processes from -3 and -20°C , respectively; the stoichiometry ratio is 2.0 corresponding to the operating current density for both the anode and cathode inlet gases; the inlet gas temperatures are equal to the ambient temperatures; the membrane thickness is $30 \mu\text{m}$; the initial water content in the membrane and catalyst layers is 6.2; the initial ice volume fraction is 0 everywhere in the cell; and the relative humidity is 0 for all the inlet gases.

inlet gases; the inlet gas temperatures are equal to the ambient temperatures; the membrane thickness is 30 μm ; the initial water content in the membrane and CLs is 6.2; the initial ice volume fraction is 0 everywhere in the cell; and the relative humidity is 0 for all the inlet gases.

It can be observed from Fig. 4 that the cell voltage evolutions are consistent between the model predictions and experimental data. For the model predictions in this figure, the current density is 0 before the cell starts and therefore the cell voltage is assumed to be equal to the reversible voltage (Eq. (38)). When the cell starts, current is drawn from the cell and the voltage loss suddenly increases, resulting in sharp decrements of the cell voltages upon load for the two cold start processes. This observation is consistent with the numerical results in [29]. It should be noticed that the current densities in the experimental measurements in [5] were increased linearly from 0 to 0.04 A cm^{-2} in the first 80 s of the cold start processes due to the convenience of operation. Therefore, the model predictions obtained by starting the cell directly with a constant current density (0.04 A cm^{-2}) are compared with the experimental data from the midpoint (40 s) of the initially linear increment of the current density to conserve the total charge capacity (the x-coordinate for the experimental data in Fig. 4 is therefore 40 s less than the measurements in [5]). The same method was also used in [27] to compare the results from the model predictions with the experimental data. Due to the neglect of the initially linear increment of the current density, the cell voltages from the experimental measurements for the two cold start processes are all higher than the model predictions at the beginning (10 s). After the first 40 s (the initially linear increment of the current density is over), the model predictions agree quite well with the experimental data for the successful cold start process (from -3°C). For the failed cold start process (from -20°C), the model predictions agree well with the experimental data in the time period from 40 to 110 s. After that, the cell voltage starts decreasing due to the ice formation, and the model predictions show a sharper decrement of the cell voltage than the experimental data. Since the present model assumes a linear relationship to reflect the blockage of the reaction sites due to the ice and liquid water formation [27–29] (Eqs. (7) and (8)), further modifications of this linear relationship may result in better agreements with the experimental data. However, further pore-scale investigation on the ice and liquid water distributions are needed to find better correlations.

3.2. Effects of inlet gas temperatures and startup current densities

Simulations for the cell starting from -10 and -3°C are carried out in this study. Previous experimental measurements showed that it is possible to start the cell successfully from -3°C and the startup from -10°C may be failed [5]. In fact, based on the cell properties and operating conditions for this study (listed in Table 1), it is also found that the cell can be successfully started from -3°C but failed from -10°C . However, it should be noticed that the cold start capability mainly depends on the cell properties and operating conditions, as successful cold start processes from -10°C were also reported in the simulation studies [29,50] with different cell properties and operating conditions. The results in [50] showed that starting the cell with constant cell voltages could lead to faster cold start processes than starting the cell with constant current densities, therefore the numerical simulations in this paper focus on the cold start processes with a constant cell voltage of 0.3 V. Such a low cell voltage is considered in order to produce high startup current densities considered to be beneficial for a successful cold start process [26]. Based on the measurements of the water contents and distributions after purge in [51], as well as the setup of the initial conditions in [29], the initial water content is set to be 5.0 in the membrane and CLs,

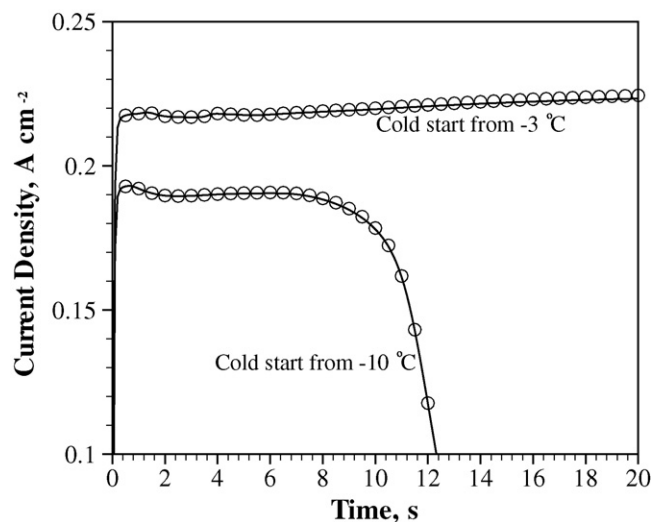


Fig. 5. Effects of the inlet gas temperatures on the cold start performance for the two different cold start processes from -3 and -10°C , respectively (solid line: the inlet gas temperatures are set in the simulations equal to the ambient temperatures; symbol $\circ$: the inlet gas temperatures are set in the simulations equal to the cell temperatures).

and the initial ice volume fraction is set to be 0.15 in the CLs and GDLs.

As mentioned earlier, since the numerical simulations in this paper only consider a straight cell (part of a whole cell), and for different locations of the straight cell in the whole cell, the inlet gas temperatures of the straight cell can be equal to the ambient temperature or the cell temperature. Fig. 5 shows the comparison between the two inlet gas conditions for the two cold start processes. It can be noticed that these two inlet gas conditions result in almost the same cold start performance, due to the fact that the extra heat carried by the inlet gases (the case that the inlet gas temperatures equal to the cell temperatures) is much lower than the heat generation from the electrochemical reaction. For the successful cold start (from -3°C), the case with the inlet gas temperatures equal to the cell temperatures produces higher and higher current densities than the case with inlet gas temperatures equal to the ambient temperature, due to the continuously increased cell temperature during the successful cold start process.

It can also be noticed that the cold start processes in Fig. 5 are much faster than in Fig. 4, due to the fact that the startup current densities in Fig. 5 (around 0.2 A cm^{-2}) is much higher than in Fig. 4 (0.04 A cm^{-2}). For a lower startup current density, the heat and water generation rates are also lower, and with a lower water generation rate the ice formation rate is lower as well, the membrane and gas streams can also take larger amounts of water (since the cold start process is longer) which further extends the duration of the cold start process. Significant difference in the durations of the cold start processes with different startup current densities can also be found in the simulations studies [50].

Since no noticeable difference between the two inlet gas conditions is observed, the rest of the investigations in this paper only focus on the condition that the inlet gas temperatures equal to the cell temperatures. The details of the operating conditions are listed in Table 1. Both the failed (from -10°C) and successful (from -3°C) cold start processes are investigated in the next two sections.

3.3. Failed cold start (from -10°C)

The evolutions of both the current density and ice volume fraction in cathode CL for the cold start from -10°C are shown in Fig. 6. The current density increases quickly at the beginning due

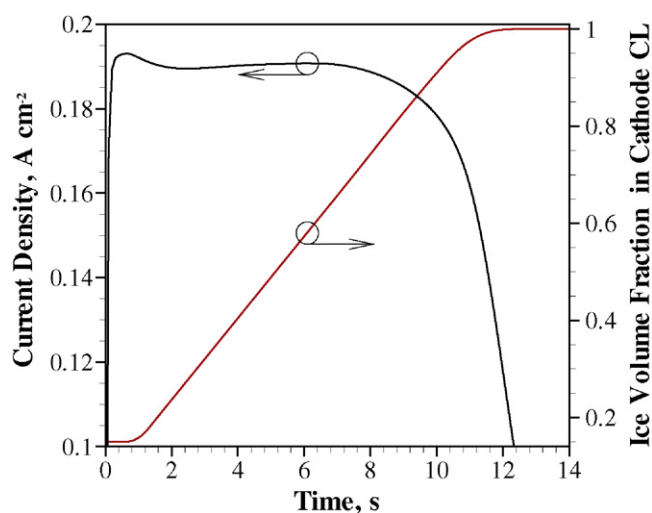


Fig. 6. Evolutions of current density and ice volume fraction in the cathode catalyst layer for the cold start from -10°C .

to the fast electrochemical double layer charging and discharging process. Then the variation of the current density becomes the minimum for a period of time, owing to the combined effects of the membrane hydration, temperature increment and ice blockage. Finally the current density drops fast indicating that the cold start process is failed. This trend is consistent with the experimental measurements in [52]. The fast drop of the current density is caused by the ice blockage on the active catalyst surface, and it occurs when the ice volume fraction in the cathode CL reaches unity (higher than 0.9). For the cold start with constant current densities, the variations of the cell voltages in both the numerical

[24,27] and experimental [4,5] results showed very similar trends, and it was reported that the cell voltage starts dropping fast when the ice volume fraction in cathode CL reaches unity [24]. The linear increment of the ice volume fraction shown in Fig. 6 is also consistent with the numerical results in [24]. As mentioned in Section 2.2, the product water is assumed to be in the ionomer in cathode CL. During the cold start process, the mass transfer between the water in the ionomer and water vapour takes place, and the freezing of the water in the ionomer into ice occurs as well, as described by the equations shown in Table 7. Therefore, there are two sources of ice formation: desublimation of vapour and freezing of membrane water, as shown in Fig. 2. Before these ice formation processes can take place, the water content in the ionomer must reach certain values: to reach 14.0 in order to have saturated water vapour to desublimates (Eq. (20)), and to reach λ_{sat} (10.1 at -10°C) in order to have the membrane water to freeze (Eq. (22)). The initial water content is 5.0, therefore the ice volume fraction in the cathode CL in Fig. 6 remains unchanged in the first second of the cold start process. Such phenomenon can also be confirmed by the experimental and numerical studies in [4,5,24,27–29], and it indicates that purging the cell reduces the rate of ice formation during cold start.

The contours of the ice volume fraction in the cathode CL in the y - z plane ($x=0.025\text{ m}$, middle of the cell) for the cold start from -10°C are shown in Fig. 7. The ice first appears under the land because the temperature is low in those areas (close to the surrounding walls), which result in low λ_{sat} . In the areas close to the membrane, more water in the ionomer could diffuse into the membrane due to the large gradient of water content, and in the areas close to the flow channel, more water in the ionomer can be taken by the dry air flow. Therefore, the ice is also first generated at the locations away from both the membrane/CL and GDL/CL interfaces. As time passes, more ice is formed and the cathode CL is fully blocked (12 s), indicating that no further electrochemical reaction can take

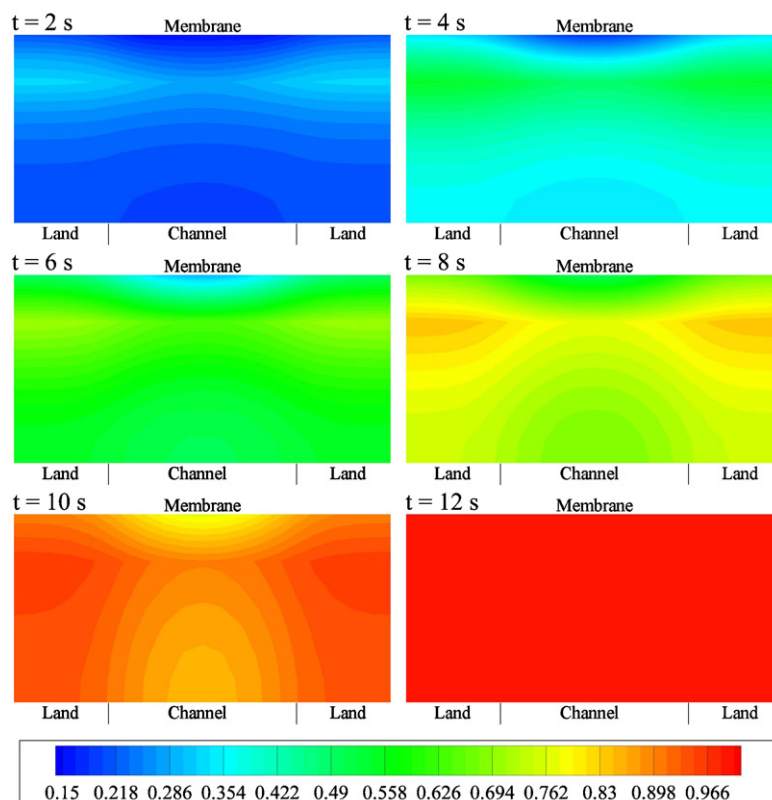


Fig. 7. Evolution of ice volume fraction in the cathode catalyst layer in the y - z plane ($x=0.025\text{ m}$, middle of the cell) for the cold start from -10°C .

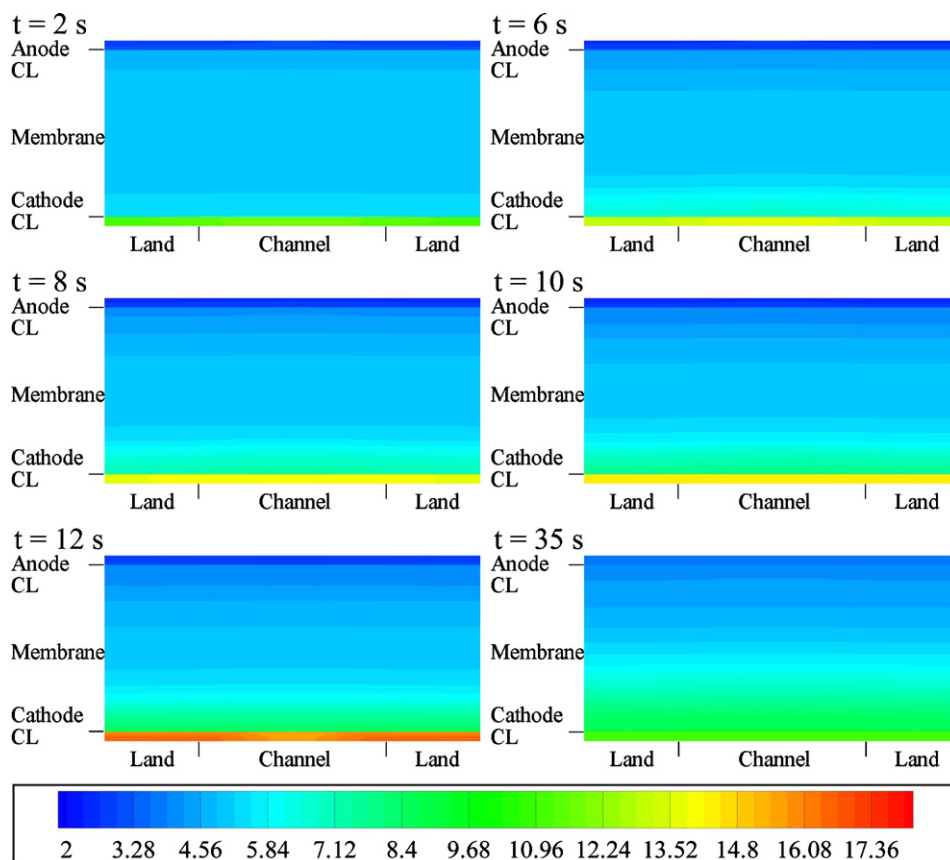


Fig. 8. Evolution of non-frozen water content in the membrane and catalyst layers in the y - z plane ($x = 0.025$ m, middle of the cell) for the cold start from -10 °C.

place. The evolution of the non-frozen water content in the membrane and CLs for the cold start from -10 °C is illustrated in Fig. 8. It can be observed that the water content in the cathode CL increases fast and the changes of the water content in the membrane and anode CL are relatively slow. The reason is that the water production rate is higher than the water diffusion rate in the ionomer (low diffusivity at subzero temperatures). Fig. 8 shows that the water content difference across the membrane is still very significant even at 35 s (23 s after the electrochemical reaction stops). Therefore, it can be concluded that increasing the ionomer fraction in the cathode CL has more significant effects than increasing the thickness of the membrane layer in reducing the amount of ice formation.

The evolutions of various heat generation rates, heat loss rate through the surrounding walls, and cell temperature for the cold start from -10 °C are presented in Fig. 9. It should be noticed that the cell temperature presented in this figure is the volume averaged temperature of the whole computational domain, and the heat generation rates and the heat loss rate are the total values for the whole computational domain. The largest heating source is found to be the ohmic heat, mainly due to the low membrane conductivity at subzero temperatures. The activation heat in this case is about half of the ohmic heat, however, the numerical simulations in [29] showed that the activation heat is higher than the ohmic heat. The reason is that the startup current density in this study (0.19 A cm^{-2}) is higher than the current density in [29] (0.1 A cm^{-2}). The last equation in Table 3 shows that the ohmic heat increases with an exponent of 2 with the increment of current density, and the activation heat changes almost linearly with the current density (the major change of η_{act} occurs at very low current densities). Both the reversible heat and latent heat are much lower, and at 12 s, all the heat generations drop down to almost zero due

to the blockage of ice. The evolutions of the heat loss through the surrounding walls and the cell temperature are in the same trend (the heat loss increases linearly with the cell temperature if the heat transfer coefficient is constant), and it should be noticed that the heat exchange through the flow inlets and outlets can be safely neglected (the level of 10^{-5} W) because the inlet gases are set to be equal to the cell temperature. The cell temperature reaches to about -3.5 °C (12 s, when the electrochemical reaction is stopped) and then decreases again due to the heat loss to the surroundings.

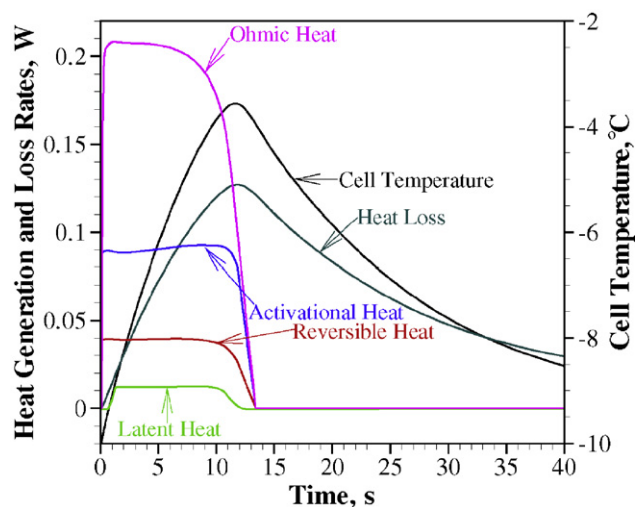


Fig. 9. Evolutions of various heat generation rates, heat loss rate through the surrounding walls, and cell temperature for the cold start from -10 °C.

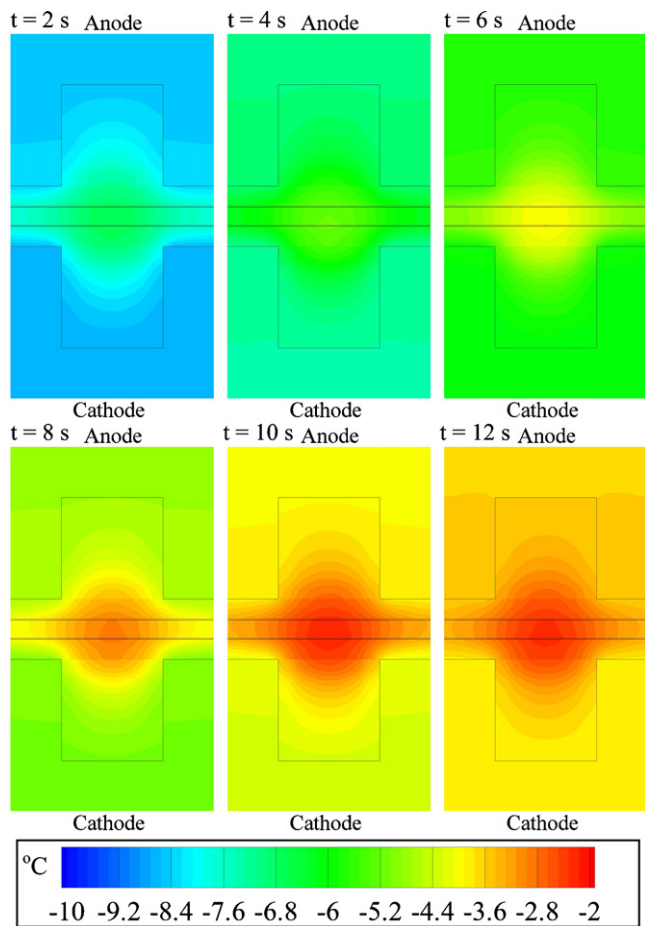


Fig. 10. Evolution of temperature in the y - z plane ($x = 0.025$ m, middle of the cell) for the cold start from -10°C .

The contours of temperature in the y - z plane ($x = 0.025$ m, middle of the cell) for the cold start from -10°C is given in Fig. 10. Due to the fact that the ohmic heat is mainly caused by the low membrane conductivity, therefore the highest temperature is very close to the membrane, a slight shift to the cathode CL due to the other heating sources (activation heat, reversible heat and latent heat). The temperature is also the highest under the flow channel rather than under the land because the heat is lost at the surrounding walls. It is also observed that the temperature in the anode is slightly higher than in the cathode, and the reasons are: (1) the ohmic heat is the highest in the anode CL (a low water content results in a low membrane conductivity); (2) the heat transfer rate is higher in the anode, due to the higher thermal conductivity of hydrogen by comparing with oxygen (listed in Table 4) and the blockage of ice in the cathode CL.

3.4. Successful cold start (from -3°C)

Fig. 11 shows evolutions of current density and cell temperature for the cold start from -3°C . In this successful cold start process, the current density increases with the increment of the cell temperature, and the increment of the cell temperature becomes the lowest at around 0°C , because heat is needed for ice melting. Even Fig. 11 shows that the cell temperature reaches 0°C at about 5 s, however, the evolutions of ice and liquid water volumes fractions in Fig. 12 shows that the ice starts melting at about 1 s in the CLs and 2.5 s in the GDLs, due to the higher temperatures in these regions as well as the FPD effect. The liquid water volume fractions change more dramatically than the ice volume fractions. In the cathode

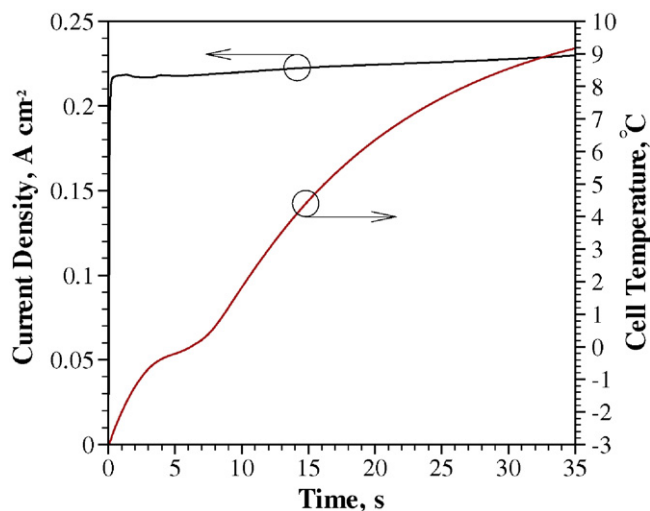


Fig. 11. Evolutions of current density and cell temperature for the cold start from -3°C .

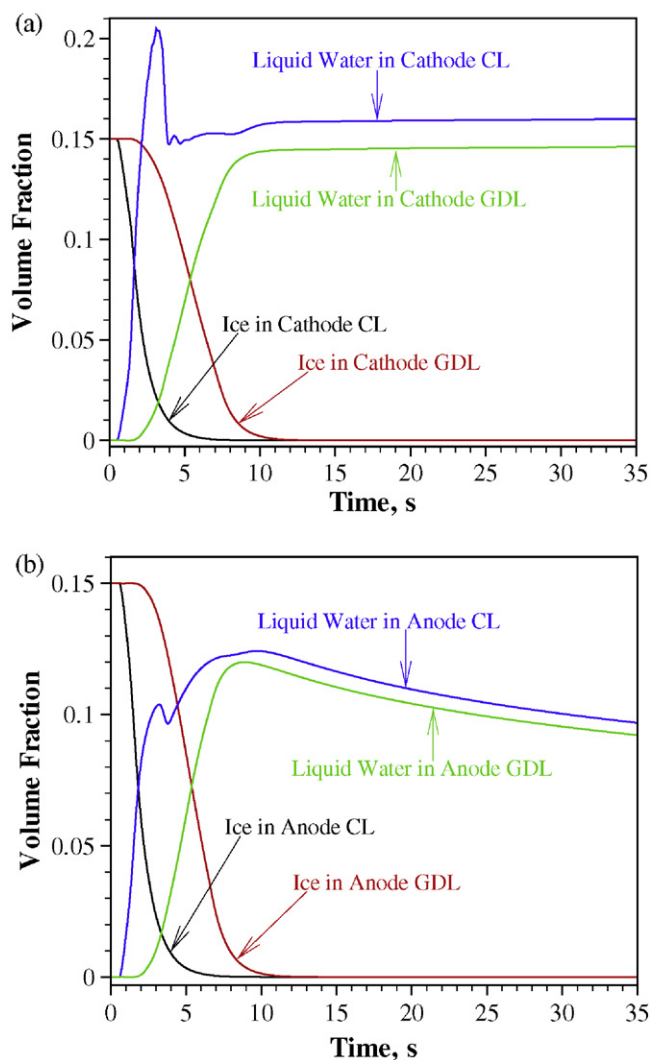


Fig. 12. Evolutions of ice and liquid water volume fractions in the catalyst layers and gas diffusion layers for the cold start from -3°C (a: cathode; b: anode).

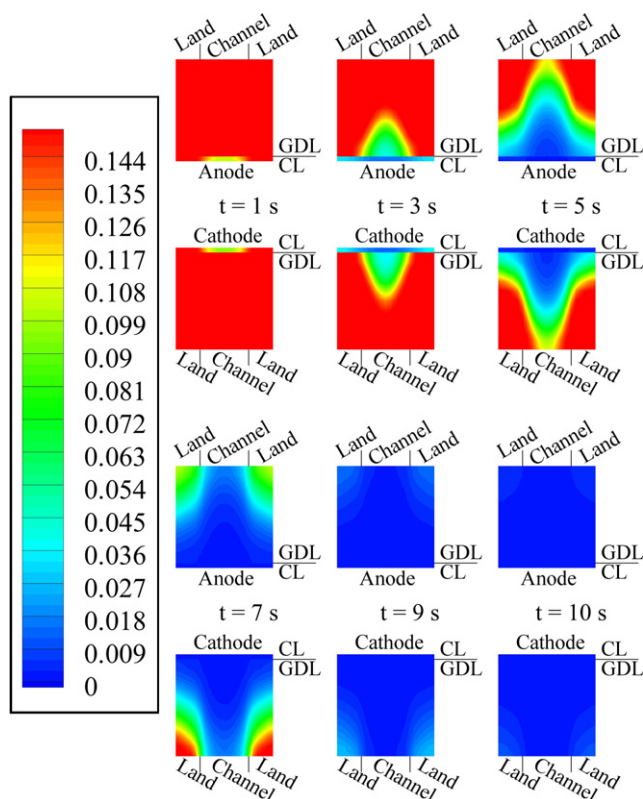


Fig. 13. Evolution of ice volume fraction in the catalyst layers and gas diffusion layers in the y - z plane ($x=0.025$ m, middle of the cell) for the cold start from -3°C .

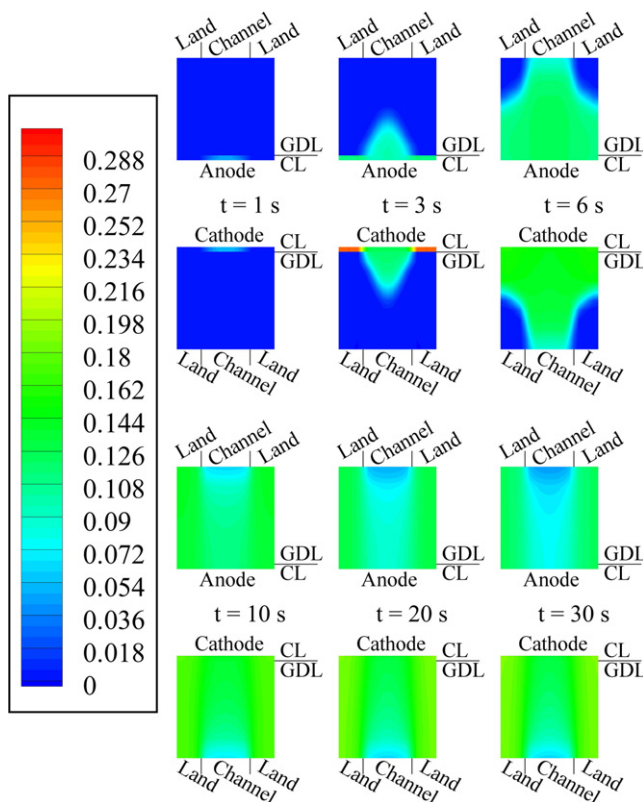


Fig. 14. Evolution of liquid water volume fraction in the catalyst layers and gas diffusion layers in the y - z plane ($x=0.025$ m, middle of the cell) for the cold start from -3°C .

(Fig. 12(a)), the liquid water volume fraction in the CL increases up to 0.2 at the beginning, due to both the ice melting and water production, followed by a fast drop due to the liquid water removal and the stoppage of the ice melting. The liquid water volume fractions in the cathode CL and GDL reach almost constants at about 10 s by balancing both the water production and removal. The liquid water volume fractions in the anode (Fig. 12(b)) are less than in the cathode, and the maximum volume fractions are reached at the end of the ice melting process, and then decrease with time due to both the water evaporation and the water absorption by the ionomer.

Fig. 13 shows the contours of ice volume fraction in the CLs and GDLs in the y - z plane ($x=0.025$ m, middle of the cell) for the cold start from -3°C . The ice first melts in the CLs under the land ($t=1$ s), then the whole CL and the GDL under the land, and finally the whole area. It is also observed that the ice melting in the anode is slightly faster than in the cathode, due to the faster heat transfer in the anode, as mentioned before. The evolution of the ice volume fraction shown in Fig. 13 matches the temperature distributions shown in Fig. 10. The contours of liquid water volume fraction in the CLs and GDLs in the y - z plane ($x=0.025$ m, middle of the cell) for the cold start from -3°C is presented in Fig. 14, and the locations of the liquid water formation matches the locations of the ice melting in Fig. 13. It should be noticed that the liquid water in the cathode CL becomes the maximum at 3 s, and at 30 s, the liquid water in the anode is still decreasing and it remains almost unchanged in the cathode, this observation is consistent with the results shown in Fig. 12.

4. Conclusion

In this paper, a mathematical model has been developed to simulate the cold start processes in a PEMFC. Unique in the present three-dimensional multiphase cold start model is the inclusion of the following phenomena: (i) the water freezing in the membrane electrolyte and its effects on the membrane conductivity; (ii) the non-equilibrium mass transfer between the water in the ionomer and the water (vapour, liquid and ice) in the pore region of the catalyst layer (CL); and (iii) both the water freezing and melting in the CL and gas diffusion layer (GDL). Good agreement between the model predictions and published experimental data has been obtained. Numerical simulations have been conducted for a single PEMFC starting at different subzero temperatures (-10°C for failed cold start and -3°C for successful cold start). Two inlet gas temperatures are considered, either equal to the cell temperatures or the ambient temperatures, and no noticeable difference in cold start performance between these two conditions are observed, due to the fact that the extra heat carried by the inlet gases for the case that the inlet gas temperatures equal to the cell temperatures is much lower than the heat generation from the electrochemical reaction. It is also found that increasing the startup current density can significantly accelerate the cold start process. For the cold start from -10°C (failed cold start), it is observed that the ice first appears under the land and at the locations away from both the membrane/CL and GDL/CL interfaces. The simulation results also show that the water production rate is higher than the water diffusion rate in the ionomer (low diffusivity at subzero temperatures), therefore increasing the ionomer fraction in the cathode CL has more significant effects than increasing the thickness of the membrane in reducing the amount of ice formation. At low cell voltages, it is found that the ohmic heat is the largest heating source during cold start, which reduces the temperature difference between the anode and cathode. The much higher conductivity of hydrogen by comparing with oxygen and nitrogen also results in a faster heating-up in the anode. For the cold start from -3°C (successful cold start), the ice melts first in the CLs and then in the GDLs, and the melt-

ing process first starts under the flow channel and then under the land. The melted water in the anode is also removed faster than in the cathode, due to both the water evaporation and the water absorption by the ionomer.

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Appendix A. Nomenclature

The financial support by the Natural Sciences and Engineering Research Council of Canada (NSERC) via a strategic Project Grant (Grant no. 350662-07) and by Auto21 is greatly appreciated.

a	water activity
A	cell geometric area (m^2)
c	molar concentration (mol m^{-3})
C_p	specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)
D	mass diffusivity ($\text{m}^2 \text{s}^{-1}$)
EW	equivalent weight of membrane ($1100 \text{ kg kmol}^{-1}$)
F	Faraday's constant ($96,487 \text{ C mol}^{-1}$)
h	latent heat (J kg^{-1}), surrounding heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
I	current density (A cm^{-2})
j	reaction rate (A m^{-3})
j_0	volumetric exchange current density (A m^{-3})
k	thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
K	permeability (m^2)
$\dot{m}$	mass flow rate (kg s^{-1})
M	molecular weight (kg kmol^{-1})
n_d	electro-osmotic drag coefficient ($\text{H}_2\text{O per H}^+$)
p	pressure (Pa)
Q	heat transfer rate (W)
r	pore radius (m)
R	universal gas constant ($8.314 \text{ J mol}^{-1} \text{K}^{-1}$)
RH	relative humidity
s	volume fraction
S	source terms, entropy ($\text{J kmol}^{-1} \text{K}^{-1}$)
t	time (s)
T	temperature (K)
T_0	volume averaged cell temperature (K)
$\vec{u}$	velocity (m s^{-1})
V	electrical potential (V)
X	mole fraction
Y	mass fraction

Greek letters

α	transfer coefficient
γ	water phase change rate (s^{-1})
ε	porosity
ζ	water transfer rate (s^{-1})
η	over potential (V)
θ	contact angle ($^\circ$)
ι	interfacial drag coefficient
κ	electrical conductivity (S m^{-1})
λ	water content in ionomer
μ	dynamic viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)
ξ	stoichiometry ratio
ρ	density (kg m^{-3})
σ	surface tension (N m^{-1})
ϕ	electrical potential (V)
ω	volume fraction of ionomer in catalyst layer

Subscripts and superscripts

a	anode
act	activation
BP	bipolar plate
c	cathode
$cell$	cell characteristic
CL	catalyst layer
$cond$	condensation
$desb$	desublimation
eff	effective
ele	electronic
EOD	electro-osmotic drag
$equil$	equilibrium
$evap$	evaporation
f	frozen
fl	fluid phase
FPD	freezing point depression
fmw	frozen membrane water
g	gas phase
GDL	gas diffusion layer
H_2	hydrogen
i, j	the i th and j th components
ice	ice
in	inlet
ion	ionic
lq	liquid water
m	mass (for source term)
mem	membrane
N	normal condition
nf	non-frozen
nmw	non-frozen membrane water
O_2	oxygen
out	outlet
ref	reference state
rev	reversible
sat	saturation
sl	solid phase including the membrane electrolyte and ice
sld	solid phase excluding the membrane electrolyte and ice
$surr$	surroundings
T	energy (for source term)
u	momentum (for source term)
vp	water vapour
$wall$	surrounding wall of the cell
0	intrinsic value
$l-i$	liquid water to ice (vice versa)
$n-f$	non-frozen membrane water to frozen membrane water (vice versa)
$n-i$	non-frozen membrane water to ice
$n-v$	non-frozen membrane water to vapour (vice versa)
$v-i$	vapour to ice
$v-l$	vapour to water liquid (vice versa)

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